

M13.8 Instruments 345-406 — Full Set

275 questions · Level 1: 128 · Level 2: 125 · Level 3: 22

Grouped by subject, then by level. Three answers each, one correct, in the Part-66 exam format.
No calculations, no worked examples from the manual, nothing aircraft-type specific.

Classification

Level 1 — recall

1. To which two subgroups can every instrument of the classical cockpit be assigned
 - a. Airframe monitoring and aircraft monitoring
 - b. Flight monitoring and engine monitoring
 - c. Flight monitoring and aircraft monitoring
2. To which group does the artificial horizon indicator ADI belong
 - a. Airframe monitoring
 - b. Flight monitoring
 - c. Aircraft monitoring
3. To which group does an RPM indicator belong
 - a. Standby devices
 - b. Aircraft monitoring
 - c. Flight monitoring
4. Which of these instruments belongs to the flight monitoring group
 - a. Fuel quantity indicator
 - b. Flap position indicator
 - c. Turn and slip indicator
5. Which devices are the airframe monitoring devices of a classical cockpit
 - a. Position indicators and RPM indicators
 - b. Pressure indicators and fuel indicators
 - c. Position indicators and pressure indicators
6. Which devices are the engine monitoring devices of a classical cockpit
 - a. RPM, exhaust gas temperature and position indicators
 - b. Position, exhaust gas temperature and fuel indicators
 - c. RPM, exhaust gas temperature and fuel indicators
7. What does the abbreviation EIS stand for
 - a. Engine Indicating System
 - b. Electronic Instrument System
 - c. Electronic Indicating System
8. What does the abbreviation EFIS stand for
 - a. Electronic Flight Indicating System
 - b. Electronic Flight Instrument System
 - c. Electronic Flight Information Source
9. Which instruments are assigned to the EFIS in the modern cockpit
 - a. The engine monitoring instruments
 - b. The airframe monitoring instruments
 - c. The flight monitoring instruments
10. What does the abbreviation EICAS stand for
 - a. Engine Indicating and Centralized Alerting System
 - b. Engine Indicating and Crew Alerting System
 - c. Electronic Indicating and Crew Alerting System
11. What does the abbreviation ECAM stand for
 - a. Electronic Centralized Aircraft Monitoring
 - b. Engine Centralized Aircraft Monitoring
 - c. Electronic Centralized Airframe Monitoring
12. What does the abbreviation CMS stand for
 - a. Central Maintenance System
 - b. Centralized Monitoring System
 - c. Central Management System
13. Which devices make up the standby set of a modern cockpit
 - a. Altimeter, variometer, horizon and compass
 - b. Altimeter, airspeed indicator, machmeter and compass
 - c. Altimeter, airspeed indicator, horizon and compass

14. A change in which of these values causes an instrument to operate
- a. Solar radiation
 - b. Cabin humidity
 - c. Earth's magnetism
15. Which gyro-related value is one of the values whose change causes an instrument to operate
- a. Gyro stability
 - b. Gyro rotor mass
 - c. Gyro bearing friction

Level 2 — understand and explain

16. An exhaust gas temperature indicator is fitted in a classical cockpit. Which group and subdivision does it belong to?
- a. Aircraft monitoring, as an airframe monitoring device
 - b. Aircraft monitoring, as an engine monitoring device
 - c. Flight monitoring, as a pressure monitoring device
17. Into which two kinds of devices is aircraft monitoring divided
- a. Airframe monitoring devices and engine monitoring devices
 - b. Airframe monitoring devices and standby devices
 - c. Flight monitoring devices and engine monitoring devices
18. Why do the aircraft monitoring systems of the modern cockpit carry different names such as EICAS or ECAM
- a. Because the name depends on the display size
 - b. Because the name depends on the aircraft type
 - c. Because the name depends on the flight phase
19. Which task do EICAS and ECAM perform in the modern cockpit
- a. They perform flight monitoring
 - b. They perform aircraft monitoring
 - c. They perform maintenance data recording
20. Why is the CMS counted as part of aircraft monitoring
- a. Because it delivers information to the flight crew
 - b. Because it delivers information to the maintenance personnel
 - c. Because it delivers information to the standby instruments
21. A fuel pressure indication changes during flight. Which value is the instrument responding to?
- a. Pressure of air
 - b. Pressure of fluids
 - c. Earth's magnetism

Level 3 — apply and reason

22. The electronic displays of a glass cockpit are lost in flight. Which speed and attitude information remains available from the standby set?
- a. Airspeed from the standby airspeed indicator and attitude from the standby horizon
 - b. Mach number from the standby machmeter and attitude from the standby horizon
 - c. Airspeed from the standby airspeed indicator and heading from the standby horizon

Cockpit displays and the Basic T

Level 1 — recall

23. What is provided to each pilot for flight attitude and flight monitoring
- His own set of engine and systems monitoring instruments
 - His own set of flight attitude and flight monitoring instruments
 - One set of flight attitude and flight monitoring instruments shared by both pilots
24. What does the Basic T set of flight monitoring instruments tell the pilot about
- Aircraft airspeed, altitude and fuel quantity
 - Aircraft attitude, airspeed and altitude
 - Aircraft attitude, heading and engine thrust
25. Which instrument of the Basic Six indicates the rate of climb or dive
- The directional gyro
 - The variometer
 - The turn and bank indicator
26. Which unit is used by the airspeed indicator of the Basic Six
- Degrees
 - Knots
 - Feet
27. An altimeter of the Basic Six set is being checked on the bench. Which unit should its scale be calibrated in?
- Knots
 - Metres
 - Feet
28. Which instrument of the Basic Six provides the compass function
- The variometer
 - The artificial horizon
 - The directional gyro
29. What does the artificial horizon of the Basic Six indicate
- Aircraft attitude
 - Rate of climb
 - Magnetic heading
30. How many essential instruments make up the set used for flying in instrument meteorological conditions
- Four
 - Six
 - Eight
31. A panel layout is described as Basic T. Which four instruments form that T?
- Airspeed indicator, attitude indicator, altimeter and vertical speed indicator
 - Attitude indicator, altimeter, compass and turn coordinator
 - Airspeed indicator, attitude indicator, altimeter and compass
32. Where is the attitude indicator located in the Basic T configuration
- On the left at the top
 - In the middle at the bottom
 - In the middle at the top
33. Which instrument is located to the right of the attitude indicator in the Basic T
- The airspeed indicator
 - The altimeter
 - The compass
34. Which instrument is located to the left of the attitude indicator in the Basic T
- The airspeed indicator
 - The vertical speed indicator
 - The altimeter
35. A panel is being rebuilt to the Basic T standard. Which instrument belongs directly below the attitude indicator?
- The airspeed indicator
 - The compass
 - The altimeter

36. Where is the magnetic standby compass often found
 - a. Beside the altimeter
 - b. Between the windscreens
 - c. Below the airspeed indicator
37. Which instruments form the lower row of the Basic T
 - a. Radio compass, directional indicator and vertical speed
 - b. Airspeed, directional indicator and vertical speed
 - c. Radio compass, directional indicator and altimeter
38. A flight instrument is listed in the wiring manual as the ADI. What does this abbreviation stand for?
 - a. Altitude Director Indicator
 - b. Attitude Director Indicator
 - c. Attitude Deviation Indicator
39. What does an ADI display
 - a. The artificial horizon and the indication from the radio altimeter
 - b. The compass and the indications from the radio navigation systems
 - c. The artificial horizon and the indication from the flight director system
40. What does HSI stand for
 - a. Horizontal Steering Indicator
 - b. Horizontal Stabiliser Indicator
 - c. Horizontal Situation Indicator
41. What is the name of the display each pilot has in an aircraft with modern displays
 - a. Primary Flight Display
 - b. Primary Flight Director
 - c. Primary Function Display
42. Which further Basic T parameter does the PFD show in addition to airspeed, artificial horizon, altitude and compass
 - a. Vertical speed
 - b. Static air temperature
 - c. Mach number
43. Where is lateral deviation presented on the primary flight display
 - a. Along the left side of the display
 - b. Along the right side of the display
 - c. Along the bottom of the display
44. Where is vertical deviation presented on the primary flight display
 - a. Along the bottom of the display
 - b. Along the right side of the display
 - c. Along the left side of the display
45. Which trend has instrumentation followed in recent years
 - a. It has been greatly simplified and duplicated
 - b. It has been greatly simplified and centralised
 - c. It has been greatly expanded and decentralised
46. Which input methods follow the integrated central control units in the trend of control element integration
 - a. Touch input, pedal input and keyboard command
 - b. Switch input, viewing direction control and stick command
 - c. Touch input, viewing direction control and voice command
47. On which display is navigation information integrated
 - a. On the E/WD
 - b. On the SD
 - c. On the ND

Level 2 — understand and explain

48. One air force fitted the same six instruments in the same panel positions in light trainers and in heavy bombers. Which advantage did this bring?
 - a. A pilot trained on one type can fly another type without an airspeed indicator
 - b. A pilot trained on one type needs no attitude instrument on another type
 - c. A pilot trained on one type can quickly get used to flying another type blind

- 49.** A pilot reads his magnetic heading and a VOR course deviation from one round instrument. Which indicator is he using?
- a.** The ADI
 - b.** The variometer
 - c.** The HSI
- 50.** A glass cockpit aircraft has no separate round flight instruments. Which statement describes the arrangement of the flight parameters?
- a.** They are laid out on the navigation display in Basic T configuration
 - b.** They are laid out on the PFD in a configuration that replaces the Basic T
 - c.** They are laid out on the PFD and correspond to the Basic T configuration
- 51.** On a glass cockpit primary flight display, which information occupies the central area?
- a.** Vertical deviation
 - b.** Attitude and guidance
 - c.** Lateral deviation
- 52.** Switches, sticks, pedals, keys and knobs were once fitted as individual operating devices. Into what does the trend in control elements integrate them?
- a.** Into integrated central control units
 - b.** Into separate air data calculators
 - c.** Into dedicated standby instruments
- 53.** On which displays is attitude information integrated in the display integration trend
- a.** On the E/WD and the SD
 - b.** On the ND and the MFD
 - c.** On the PFD and the HUD
- 54.** Engine parameters and system synoptics are being grouped onto the centre displays. Which combination applies?
- a.** Engine on the E/WD and systems on the SD
 - b.** Engine on the SD and systems on the E/WD
 - c.** Engine on the ND and systems on the E/WD

The atmosphere and ISA

Level 1 — recall

55. What is the proportion of nitrogen in dry, clean air directly above the surface of the earth
- 78%
 - 21%
 - 1%
56. What is the proportion of oxygen in dry, clean air directly above the surface of the earth
- 11%
 - 21%
 - 78%
57. Dry, clean air at the surface is made up of nitrogen, oxygen and a remaining one per cent. Which gases form that one per cent?
- Reactive gases such as hydrogen, methane and carbon monoxide
 - By-products such as water vapour, carbon dioxide and ozone
 - Noble gases such as helium, argon, neon, krypton and xenon
58. Which is the lowest layer of the atmosphere, surrounding the surface of the earth
- The stratosphere
 - The mesosphere
 - The troposphere
59. In which layer of the atmosphere do the numerous weather activities take place
- In the troposphere
 - In the stratosphere
 - In the thermosphere
60. A flight is planned across the polar region. How high does the troposphere reach there?
- About 8.5 km
 - About 16.8 km
 - About 11 km
61. An aircraft climbs through the tropopause. Which layer does it enter?
- The stratosphere
 - The mesosphere
 - The thermosphere
62. What is the name of the boundary between the troposphere and the stratosphere
- The stratopause
 - The mesopause
 - The tropopause
63. What is the approximate upper limit of the stratosphere
- About 80 km
 - About 50 km
 - About 11 km
64. Which layer of the atmosphere contains the ozone layer?
- The thermosphere
 - The stratosphere
 - The troposphere
65. Which is the outermost layer of the atmosphere
- The thermosphere
 - The mesosphere
 - The exosphere
66. An air data system is being described to a student. Which two measurements does flight monitoring require?
- Static pressure and air density
 - Dynamic pressure and air temperature
 - Static pressure and dynamic pressure
67. What is the other name for dynamic pressure
- Atmospheric pressure
 - Ram pressure
 - Barometric pressure

68. What is the other name for static pressure
- Atmospheric pressure
 - Total pressure
 - Ram pressure
69. Which flight monitoring data are calculated from static and dynamic pressure
- Barometric altitude, vertical speed, airspeed and static air temperature
 - Barometric altitude, vertical speed, airspeed and mach number
 - Barometric altitude, true airspeed, mach number and static air temperature
70. Which air density is defined at mean sea level in the international standard atmosphere
- 1.225 kg/m³
 - 0.765 kg/m³
 - 1.325 kg/m³
71. Which temperature and pressure are defined at mean sea level in the international standard atmosphere
- 15 °C and 1030 mbar
 - 15 °C and 1013 mbar
 - 5 °C and 1013 mbar
72. Which pressure does a mechanical air data instrument use to indicate barometric altitude
- Static pressure
 - Dynamic pressure
 - Total pressure
73. In which layer does the summit of Mount Everest lie
- The tropopause
 - The troposphere
 - The stratosphere
74. What is the standard atmosphere sea level pressure in the imperial system
- 29.92 mmHg
 - 29.92 inHg
 - 14.7 inHg
75. What is the standard atmosphere sea level temperature in the imperial system
- 59 °F
 - 15 °F
 - 32 °F
76. Which humidity is assumed in the standard atmosphere
- 0 %
 - 100 %
 - 50 %
77. Which air temperature is given at the tropopause in the ICAO standard atmosphere
- 43.5 °C
 - 30.5 °C
 - 56.5 °C
78. A mercury barometer is read at standard sea level conditions. Which pressure does it show?
- 1013 mm Hg
 - 29.92 mm Hg
 - 760 mm Hg
79. What is the standard atmosphere sea level pressure expressed in pounds per square inch
- 14.696 psi
 - 29.921 psi
 - 1.033 psi
80. What is the altitude of the tropopause in the international standard atmosphere
- 20 km
 - 8 km
 - 11 km

Level 2 — understand and explain

81. How does the temperature behave with increasing height in the troposphere
- It increases by 0.65 °C every 100 metres
 - It decreases by 6.5 °C every 100 metres
 - It decreases by 0.65 °C every 100 metres
82. Which is the layer above the mesopause in which the temperature increases again with height
- The mesosphere
 - The thermosphere
 - The stratosphere
83. How do static pressure, temperature and density behave as altitude increases
- All three become lower
 - Only temperature becomes lower
 - All three become higher
84. True airspeed and static air temperature are to be indicated as well. Which value must be applied as a corrective?
- Humidity
 - Temperature
 - Density
85. A pneumatic instrument is calibrated in the workshop. Which values form the basis of that calibration?
- The tropopause values of the standard atmosphere
 - The standard atmosphere values at mean sea level
 - The actual airfield values at the time of installation
86. A uniform reference value was needed for aviation. Which body introduced the international standard atmosphere?
- ICAO
 - IATA
 - EASA
87. Why has the international standard atmosphere been established
- To predict the weather conditions along a planned route
 - To set the pressure scale of the altimeter before every flight
 - To compare flight performance under greatly differing conditions

Level 3 — apply and reason

88. How does the air temperature of the standard atmosphere behave with increasing height directly above the tropopause
- It continues to fall at the same rate
 - It stops falling and stays constant
 - It begins to rise with height

Altitude, altimeter settings and RVSM

Level 1 — recall

89. What does HUD stand for
- Head Up Display
 - Heading Up Display
 - Head Unit Display
90. Which air data parameter is presented on the left-hand scale of the PFD
- Vertical speed
 - Airspeed
 - Altitude
91. Where is the reference level set on the altimeter
- On the altitude drum
 - On the pressure scale
 - On the vertical speed scale
92. What is elevation
- The altitude of an object on the ground above mean sea level
 - The altitude of the aircraft above the ground
 - The altitude of the aircraft above mean sea level
93. An airfield is listed with its height above mean sea level. Which setting code belongs to this height definition?
- QFF
 - QNE
 - QFE
94. What is height
- The altitude of the aircraft above mean sea level
 - The altitude of an object on the ground above mean sea level
 - The altitude of the aircraft above the ground
95. Which instrument normally measures height
- The air data computer
 - The low range radio altimeter
 - The barometric altimeter
96. Which setting code belongs to altitude
- QFE
 - QFF
 - QNH
97. What is set on the altimeter when QNH is used
- The current air pressure at the airfield
 - The current air pressure at mean sea level
 - The standard pressure of 1013.2 hPa
98. Which setting code belongs to the standard setting used for flight levels
- QFF
 - QNH
 - QNE
99. Which air pressure are all pilots required to set for cruise above the transition altitude
- 1030 hPa
 - 1013.2 hPa
 - 1113.2 hPa
100. An altimeter is set to the standard pressure in cruise. What is its indication called?
- True altitude
 - Pressure altitude
 - Barometric height
101. What is a flight level
- The height above the mean sea level pressure of the day
 - The height above the airfield pressure level
 - The height above the 1013.25 hPa pressure level

- 102.** What is the minimum margin by which the transition level lies above a transition altitude of 5000 ft MSL
- 1000 ft
 - 500 ft
 - 2000 ft
- 103.** An operator prepares its fleet for RVSM operations. What does this abbreviation mean?
- The halving of vertical safety distances at higher flight altitudes
 - The doubling of vertical safety distances at higher flight altitudes
 - The halving of lateral safety distances at higher flight altitudes
- 104.** Which minimum vertical separation is determined by the semicircular rule for IFR traffic up to FL 285
- 2000 ft
 - 1000 ft
 - 500 ft
- 105.** Which vertical separation was applied above FL 285 until 1997
- 1000 ft
 - 2000 ft
 - 4000 ft
- 106.** Which altitude change corresponds to a pressure change of 1 hPa at mean sea level
- 30 ft
 - 120 ft
 - 300 ft
- 107.** What is RVSM airspace
- Airspace above FL 290 with 2000 ft vertical separation
 - Airspace above FL 285 with 500 ft vertical separation
 - Airspace above FL 290 with 1000 ft vertical separation
- 108.** Which specification must an aircraft fulfil to be approved for RVSM
- MNPS
 - MMEL
 - MTBF
- 109.** Which item is listed as an RVSM-sensitive component of the aircraft
- The autopilot in altitude hold mode
 - The inertial reference unit
 - The radio altimeter
- 110.** An aircraft is being equipped for flight in RVSM airspace. How many independent altimeters must it have?
- 1
 - 2
 - 3
- 111.** RVSM was first driven by the traffic over the Atlantic. In which further regions has it since been introduced?
- Over Europe, Africa and Pacific routes
 - Over Europe, America and Pacific routes
 - Over Europe, America and polar routes
- 112.** Who must receive additional mandatory training for the introduction of RVSM
- Pilots and maintenance personnel
 - Maintenance personnel and ground handlers
 - Pilots and air traffic controllers
- 113.** The RVSM training covers corrosion of the pitot-static sensors. Which replacement interval applies to them?
- Every 6,900 hours
 - Every 960 hours
 - Every 9,600 hours
- 114.** What does SSEC stand for
- Static Source Error Computation
 - Static Source Error Correction
 - Static Sensor Error Correction
- 115.** For which previously issued navigation approval is the MNPS Approval granted
- ADC (Air Data Computer)
 - SSEC (Static Source Error Correction)
 - LRN (Long Range Navigation)

Level 2 — understand and explain

116. Where can the conversion from pressure to indication take place
- In the instrument itself or separately in the transponder
 - In the instrument itself or separately in air data calculators
 - In the pitot probe itself or separately in the instrument
117. Two aircraft show the same air data but the instruments work differently inside. On what does the way they function depend?
- On the manufacturer and the state of technological development
 - On the manufacturer and the reference pressure selected
 - On the operator and the altitude monitoring programme
118. How does the HUD present the PFD data
- So that the pilot can see it when he looks down at the centre pedestal
 - So that the pilot can see it when he looks through the windshield
 - So that the pilot can see it only when the autopilot is engaged
119. A barometric altimeter is set to QFE while the aircraft stands on the airfield. What does it indicate?
- Airfield elevation
 - Current flight level
 - Zero altitude
120. What does a barometric altimeter set to QFE indicate in flight
- The height above the airfield, as long as the air pressure does not change
 - The flight level, as long as the air pressure does not change
 - The altitude above mean sea level, as long as the air pressure does not change
121. An aircraft stands on an airfield with QNH set on its altimeter. What does the altimeter show?
- Zero altitude
 - Transition level
 - Airfield elevation
122. A crew selects QNH on both altimeters. In which phases of flight is this setting used?
- During landing and take-off, up to transition altitude and below transition level
 - During landing and take-off, above transition altitude and above transition level
 - During cruise, above transition altitude and above transition level
123. Above which altitude do all pilots set the standard pressure in cruise flight
- 5000 ft above mean sea level
 - 2000 ft above mean sea level
 - 10000 ft above mean sea level
124. All aircraft in cruise use the same standard pressure setting. Which purpose does this serve?
- It ensures correct indication of the airfield elevation
 - It ensures a correct height above the airfield
 - It ensures safe vertical separation
125. What is the transition altitude
- After take-off, the height at which the altimeter is set from STD to QNH
 - Prior to landing, the height at which the altimeter is set from QNH to STD
 - After take-off, the height at which the altimeter is set from QNH to STD
126. What is the transition level
- Prior to landing, the last usable flight level at which the altimeter is set from STD to QNH
 - After take-off, the last usable flight level at which the altimeter is set from STD to QNH
 - Prior to landing, the last usable flight level at which the altimeter is set from QNH to STD
127. What is the purpose of reducing the vertical separation minimum
- To allow aircraft to climb above the tropopause
 - To reduce the workload of the air data computers
 - To accommodate more aircraft in the same airspace
128. Why was the separation above FL 285 formerly increased
- Lack of radar coverage and reduced transponder range at high altitude
 - Lack of measurement precision and reduced altimeter sensitivity at high altitude
 - Lack of autopilot authority and reduced control response at high altitude

- 129.** What does static source error depend on
- The fixed location of the air data computer
 - The selected reference pressure on the altimeter
 - The fixed location of the static pressure inlet
- 130.** Work is planned around the static ports of an RVSM aircraft. Which condition must the structure and surroundings of the pitot-static sensors be left in?
- Smoothed with filler
 - Sealed and painted
 - Clean and clear
- 131.** What is the function of the warning device required for flight in RVSM airspace
- It indicates when the aircraft enters RVSM airspace
 - It indicates when the altimeter setting differs from 1013.2 hPa
 - It indicates when the aircraft leaves the predetermined height
- 132.** Which autopilot capability is required for flight in RVSM airspace
- It must be able to maintain the heading
 - It must be able to maintain the airspeed
 - It must be able to maintain the altitude
- 133.** Which transponder requirement applies to an aircraft flying in RVSM airspace?
- It must transmit altitude and be connected to the altitude measurement unit
 - It must transmit altitude and be connected to the radio altimeter
 - It must transmit heading and be connected to the altitude measurement unit
- 134.** Which operational advantage is gained by an RVSM approved aircraft
- It can use a reduced number of flight levels to save maintenance costs
 - It can use the increased number of flight levels to fly above the tropopause
 - It can use the increased number of flight levels to save time and fuel
- 135.** When RVSM is fully implemented, between which flight levels will the RVSM and MNPS requirements apply?
- From FL 290 to FL 410
 - From FL 290 to FL 420
 - From FL 270 to FL 430
- 136.** A fleet of six aircraft is to be cleared for RVSM. Who must apply for the Airworthiness Approval and for what?
- The aircraft operator, for each individual aircraft
 - The aircraft manufacturer, for each individual aircraft
 - The aircraft operator, for the whole fleet at once
- 137.** What must each aircraft undergo before RVSM Airworthiness Approval
- Replacement of the air data computers and the altitude monitoring programme
 - Modification according to required service bulletins and the altitude monitoring programme
 - Modification according to required service bulletins and a full static system overhaul
- 138.** The RVSM training covers surface waviness in the critical area. Which causes of that waviness are named?
- Fasteners, oxygen discharge discs and paint drops
 - Rivets, oxygen discharge discs and corrosion pits
 - Fasteners, static dischargers and paint drops
- 139.** Which detail of the air data computer must the mandatory RVSM training cover
- Its cooling airflow
 - Its part numbers
 - Its software loading procedure
- 140.** Which approval makes the aircraft eligible for the RVSM test flight
- The RVSM Airworthiness Approval
 - The MNPS Approval
 - The RVSM Operational Approval
- 141.** The altitude monitoring programme and the test flight have both been completed successfully. Which approval does the operator now receive?
- The RVSM Operational Approval
 - The RVSM Airworthiness Approval
 - The MNPS Approval

142. Which documents are issued in revised form once the RVSM conditions are fulfilled

- a.** The operation and maintenance manuals
- b.** The operation and training manuals
- c.** The maintenance and weight and balance manuals

Level 3 — apply and reason

143. An aircraft cruises at 4000 ft above mean sea level in an area with a transition altitude of 5000 ft. Which altimeter setting must be used?

- a.** QNH
- b.** QNE
- c.** QFF

144. The same pressure change of 1 hPa is worth 30 ft at sea level but 120 ft at FL 360. Which explanation applies?

- a.** Air density and pressure fall with altitude, so a given pressure step spans more altitude
- b.** Air density and pressure rise with altitude, so a given pressure step spans more altitude
- c.** Air temperature falls with altitude, so the altimeter capsule expands more

145. A dent is found in the skin close to a static port of an RVSM approved aircraft. Which consequence must be expected?

- a.** A considerable influence on the radio altimeter indication
- b.** A considerable influence on the mach number only
- c.** A considerable influence on the altitude indication

146. An altimeter fails in cruise and the aircraft loses its RVSM capability. Which action is required?

- a.** A quick exit of RVSM airspace
- b.** A change of the altimeter setting to QNH
- c.** A descent to the transition level

Airspeed

Level 1 — recall

147. What does the abbreviation IAS stand for
- a. Inertial airspeed
 - b. Increased airspeed
 - c. Indicated airspeed
148. Which airspeed is displayed in aircraft equipped with air data computers and glass cockpits
- a. The equivalent airspeed
 - b. The true airspeed
 - c. The calibrated airspeed
149. Which two pressures are supplied to the air data computer for the airspeed computation
- a. Static pressure and reference pressure
 - b. Total pressure and dynamic pressure
 - c. Total pressure and static pressure
150. For which purpose is the true airspeed needed
- a. For engine trim purposes
 - b. For navigation purposes
 - c. For design data purposes

Level 2 — understand and explain

151. An airspeed indicator with bellows is connected to the pitot and static systems. Which pressure does its indication directly follow?
- a. The static pressure as the difference between ambient and total pressure
 - b. The dynamic pressure as the difference between total and static pressure
 - c. The total pressure as the sum of dynamic and ambient pressure
152. Under which conditions does an airspeed indicator with bellows show the real speed of the aircraft
- a. At mean sea level with standard conditions
 - b. At cruise altitude with standard conditions
 - c. At mean sea level with compressibility corrected
153. An air data computer converts the indicated airspeed into the calibrated airspeed. Which two errors does it remove?
- a. The error from the indicator and the error from the static pressure source
 - b. The error from the pitot source and the error from air density
 - c. The error from the indicator and the error from air compressibility
154. A mechanical standby airspeed indicator with bellows is fed with total and static pressure. Which airspeed does it display?
- a. The calibrated airspeed
 - b. The equivalent airspeed
 - c. The indicated airspeed
155. Which unit performs the correction of instrument error and static source error
- a. The air data computer
 - b. The pitot probe
 - c. The airspeed indicator
156. Under which condition is the speed formula valid
- a. Only when the medium used is incompressible
 - b. Only when the medium used is at standard temperature
 - c. Only when the medium used is above the tropopause
157. An aircraft accelerates to a high speed and the total pressure rises more than the speed formula predicts. What produces this additional pressure?
- a. Air compresses at higher speeds and generates a total pressure increase
 - b. Air expands at higher speeds and generates a total pressure decrease
 - c. Air compresses at lower speeds and generates a static pressure increase
158. How does the compressibility error change with airspeed and altitude
- a. The higher the airspeed and altitude, the larger the error
 - b. The higher the airspeed and altitude, the smaller the error
 - c. The lower the airspeed and the higher the altitude, the larger the error

- 159.** What is the equivalent airspeed
- The calibrated airspeed after the compressibility error has been removed
 - The true airspeed after the density calculation has been removed
 - The indicated airspeed after the instrument error has been removed
- 160.** Why is the equivalent airspeed not shown to the crew of a commercial airplane
- It is only valid below the tropopause in standard conditions
 - It is only important for design data and further calculations
 - It is only available when the air data computer is in test
- 161.** A navigation computation needs the speed of the aircraft through the surrounding air mass. Which airspeed is this?
- The calibrated airspeed
 - The equivalent airspeed
 - The true airspeed
- 162.** In which order are the airspeeds derived from the pressure inputs
- IAS, CAS, EAS, TAS
 - CAS, IAS, TAS, EAS
 - IAS, EAS, CAS, TAS
- 163.** Which calculation is made between EAS and TAS
- A static source correction
 - A density calculation
 - A compressibility correction
- 164.** Which correction is applied between CAS and EAS
- Correction of compressibility error
 - Correction of instrument error
 - Correction of static source error
- 165.** The compressibility error for a flight condition is known in knots. What is done with this value to obtain the equivalent airspeed?
- It is subtracted from the calibrated airspeed
 - It is subtracted from the true airspeed
 - It is added to the calibrated airspeed

Level 3 — apply and reason

- 166.** Why is the equivalent airspeed lower than the calibrated airspeed at high speed and altitude?
- Because compression of the air lowers the static pressure, so the indication is too low
 - Because reduced density lowers the total pressure, so the indication is too high
 - Because compression of the air raises the total pressure, so the indication is too high
- 167.** Two aircraft fly at the same calibrated airspeed, one at 20,000 ft and one at 40,000 ft. Which one has the larger compressibility error?
- The aircraft at 20,000 ft
 - The aircraft at 40,000 ft
 - Both have the same error

Barometers, capsules and instrument errors

Level 1 — recall

168. How is the tube of a liquid-based barometer arranged
- Vertical and hermetically sealed at the bottom
 - Horizontal and hermetically sealed at both ends
 - Vertical and hermetically sealed at the top
169. Which height does mercury reach in a mercury barometer under normal conditions
- 760 mm
 - 1013 mm
 - 560 mm
170. What forms the sensing element of an aneroid barometer
- A small capsule made of thin metal
 - A small tube filled with thin fluid
 - A small chamber made of thin glass
171. What is the residual pressure inside an aneroid capsule
- About 1013 mbar
 - About 50 mbar
 - About 5 mbar
172. How many capsules are used at most in the stack of an improved barometer
- Up to eight
 - Up to four
 - Up to sixteen
173. What is the function of a Bourdon tube
- It converts pressure into a proportional temperature signal
 - It converts pressure into a proportional electrical signal
 - It converts vibration into a proportional electrical signal
174. Which cross-sectional shape does the curved tube of a Bourdon transducer have
- A round shape
 - An oval shape
 - A square shape
175. Which device is usually used as the position sensor of a Bourdon tube
- A vibration pickup
 - A synchro transmitter
 - A resolver winding
176. Which temperature range must a pressure sensor flight instrument cover
- 40 °C to +55 °C
 - 50 °C to +70 °C
 - 70 °C to +50 °C
177. How many revolutions must the altimeter needle make between -1,000 feet and +50,000 feet
- 15 revolutions
 - 50 revolutions
 - 51 revolutions
178. Which indication is given by the variometer of a pressure instrument panel
- The vertical speed
 - The Mach number
 - The barometric altitude
179. An instrument has to be connected to the aircraft pitot and static system. How is the pressure information brought to it?
- Via wiring and pressure transducers with screw or quick release connectors
 - Via piping and flexible tubes with screw or quick release connectors
 - Via piping and rigid tubes with soldered or welded connectors
180. Which stroke do diaphragm capsules supply as the measurable value
- 4 to 5 mm
 - 14 to 15 mm
 - 0.4 to 0.5 mm

Level 2 — understand and explain

- 181.** What creates the negative pressure at the top of a liquid barometer tube
- The ambient air pressure acting on the container
 - The fluid flowing out of the tube due to its own weight
 - The expansion of the glass tube with temperature
- 182.** The mercury column of a barometer stops moving at a certain height. Which pressure balances the outflow of the fluid?
- The negative pressure counteracting the ambient air pressure
 - The ambient air pressure counteracting the outflow of the fluid
 - The temperature expansion of the fluid in the container
- 183.** Which effect must be taken into account when reading a mercury barometer
- Both the mercury and the glass tube expand with increasing temperature
 - Only the mercury expands while the glass tube stays unchanged
 - Both the mercury and the glass tube contract with increasing temperature
- 184.** A mercury barometer is read in a workshop that is warmer than standard. What must be done to obtain an exact result?
- The reading must be adjusted to the residual capsule pressure
 - The reading must be adjusted to the local field elevation
 - The reading must be adjusted to standard conditions
- 185.** The capsule of an aneroid barometer holds a residual pressure of about 5 mbar. Which effect does this residual pressure compensate?
- It compensates for changes in the friction of the mechanism caused by wear
 - It compensates for changes in the density of the air caused by altitude
 - It compensates for changes in the elasticity of the metal caused by temperature
- 186.** A barograph is built with a stack of capsules instead of a single one. Why is this done?
- To increase the residual pressure of the capsules
 - To increase the reliability of measurements
 - To increase the temperature range of the instrument
- 187.** How does an aneroid capsule react to increasing air pressure
- It is expanded
 - It is unchanged
 - It is compressed
- 188.** The aneroid capsule of a barometer changes its shape with pressure. How does this movement reach the indication?
- It is transferred electrically to an indicating needle
 - It is transferred mechanically to an indicating needle
 - It is transferred hydraulically to an indicating needle
- 189.** Which pressure is applied to the outside of the capsule in an aneroid altimeter
- The static pressure
 - The dynamic pressure
 - The total pressure
- 190.** Which additional provision, beyond the pressure capsule and the indicating linkage, is built into an aneroid altimeter
- A compensation for the effect of humidity
 - A compensation for the effect of acceleration
 - A compensation for the effect of temperature
- 191.** A Bourdon tube transducer is opened for inspection. Which two parts carry out the pressure conversion?
- The curved tube and a drive coil
 - The curved tube and a position sensor
 - The straight tube and a position sensor
- 192.** A Bourdon tube transducer is fitted with a synchro transmitter. Which quantity does this sensor convert into an electrical signal?
- The position of the tube tip
 - The temperature of the tube wall
 - The pressure inside the tube
- 193.** How is an applied pressure distributed inside a Bourdon tube
- Only at the sealed tube tip
 - Equally throughout the whole tube
 - Only at the oval section of the tube

- 194.** What does the applied pressure try to do to the cross section of a Bourdon tube
- Change it from oval to rectangular
 - Change it from circular to oval
 - Change it from oval to circular
- 195.** Pressure is applied to a Bourdon tube and its oval section is pushed towards a circular one. How does the tube react?
- A straightening of the tube that moves the tip outwards
 - A further curving of the tube that moves the tip inwards
 - A lengthening of the tube that leaves the tip where it is
- 196.** What is the amount of tube tip movement proportional to
- The tube temperature
 - The signal frequency
 - The applied pressure
- 197.** Which property is required from a pressure sensor flight instrument
- Insensitivity to acceleration
 - Sensitivity to acceleration
 - Insensitivity to dynamic pressure
- 198.** Why is there little leeway for manufacturers of pressure sensor instruments
- Most of the requirements are defined and required by law
 - Most of the requirements are defined by the airline operator
 - Most of the requirements are defined by the manufacturers
- 199.** Which instruments still use pressure sensors today
- Only engine indications
 - Only primary flight displays
 - Only back-up instruments
- 200.** Several thin metal diaphragm capsules are stacked into a single unit in a pressure sensor instrument. What is this assembly called?
- A bellows
 - A torsion bar
 - A resonator
- 201.** An instrument panel is supplied only from the aircraft pitot and static system. Which group of instruments is this?
- Airspeed indicator, altimeter, machmeter and variometer
 - Airspeed indicator, altimeter, machmeter and attitude indicator
 - Airspeed indicator, altimeter, variometer and radio altimeter
- 202.** A diaphragm capsule delivers only a stroke of 4 to 5 mm with a small force. What must the transfer mechanism do with it?
- Convert it into a damped rotation of the needle
 - Convert it into an amplified rotation of the needle
 - Convert it into an amplified stroke of the capsule
- 203.** Which effect does friction have in the transfer mechanism of an aneroid instrument
- It raises the capsule hysteresis
 - It damps the capsule stroke
 - It falsifies the measured result
- 204.** How is friction in the indicator mechanism kept to a minimum
- With suitable bearings and clean construction of all moving parts
 - With sealed bearings and grease filling of all moving parts
 - With heavier levers and increased spring tension on all moving parts
- 205.** In the altimeter one further measure helps to reduce the effect of friction. Which measure is this?
- Damping of the capsule
 - Vibration of the instrument
 - Balancing of the needle
- 206.** Play in the gearing between the pinion and the tooth segment would falsify the indication. How is this play taken out?
- By spring tension on one side
 - By increased bearing clearance
 - By spring tension on both sides
- 207.** What is meant by the hysteresis of a membrane capsule
- The stroke lags behind the pressure when a constant pressure acts for a long time
 - The curve with increasing pressure differs from the curve with decreasing pressure
 - The curve with increasing pressure differs from the curve with increasing temperature

- 208.** What causes the elastic aftereffect of a capsule
- A constant pressure acting on the capsule over a longer period of time
 - A temperature change acting on the capsule mounting
 - A rapidly changing pressure acting on the capsule during a climb
- 209.** What is the result of the slow yielding of a capsule under constant pressure
- A permanent offset of the needle
 - A change of the residual pressure
 - A time lag in the stroke
- 210.** Friction in the transfer mechanism, hysteresis, elastic aftereffect and temperature all work against precise operation. Which instruments do these error influences affect?
- Only metric scale altimeters
 - All aneroid type instruments
 - Only digital air data computers
- 211.** Which part provides the temperature compensation in the mechanical altimeter mechanism
- A second diaphragm capsule
 - A spring tensioned pinion
 - A bimetal element
- 212.** The hysteresis and the elastic aftereffect of a capsule are shown as characteristic curves. Which quantity of the capsule is plotted against the applied pressure?
- The diaphragm stroke
 - The residual capsule pressure
 - The needle friction
- 213.** Which part of the altimeter mechanism drives the indicating needle directly
- The steel pins in the capsule stack
 - The bimetal rod on the bellows mounting
 - The pinion on the pointer axis

Level 3 — apply and reason

- 214.** An altimeter needle must turn 51 times over its full range. Which demand does this place on the instrument?
- The highest demands on case sealing and static line length
 - The highest demands on electrical power and signal bandwidth
 - The highest demands on material and fine mechanical precision
- 215.** Which measures are used to characterize the properties of a capsule for its intended use
- Special coatings, artificial ageing and particular bearing materials
 - Special alloys, natural ageing and particular residual pressures
 - Special alloys, artificial ageing and particular waveforms
- 216.** Which parts make temperature compensation demanding over the range -70 to +50 degrees C
- The capsule mounting and the flexible tubes in the static system
 - The pinion bearings and the quick release connectors of the instrument
 - The capsule mounting and the variable steering levers in the indicator mechanism
- 217.** An altimeter is held at one steady pressure on the test bench for a long time and the indication slowly drifts. Which error influence explains this?
- The hysteresis of the capsule
 - The backlash of the pinion gearing
 - The elastic aftereffect of the capsule
- 218.** An altimeter reads slightly differently at the same test pressure depending on whether the pressure was raised or lowered to reach it. Which error influence is shown?
- Hysteresis of the diaphragm capsule
 - Friction in the transfer mechanism
 - Elastic aftereffect of the diaphragm capsule
- 219.** Through which parts must the smallest pressure change act to deflect the needle exactly
- From the capsule via bellows, resonator and coil
 - From the capsule via levers, shaft and cogwheel
 - From the diaphragm via strain gauges, bridge and amplifier

Pressure transducers

Level 1 — recall

220. What is the function of a pressure transducer
- To convert electrical signals into measured pressures
 - To convert measured pressures into mechanical needle movement
 - To convert measured pressures into electrical signals
221. In which equipment are pressure transducers used
- In flight control computers
 - In inertial reference units
 - In air data computers
222. Which type of output signal does the latest generation of pressure transducers deliver
- A synchro signal
 - An analog signal
 - A digital signal
223. A modern pressure transducer needs a fixed reference for its measurement. Which reference value is used?
- An ambient pressure or 5 mbar
 - A vacuum or 1013 mbar
 - A vacuum or 760 mbar
224. Which output signal is produced by a pressure cell transducer with a resolver
- A frequency signal
 - An analog signal
 - A digital signal
225. A force compensated pressure cell is opened. Which two electrical conversion elements are found in it?
- An oscillator and a resonator
 - A resolver and an E-pickup
 - A torsion bar and an E-pickup
226. Which elements perform the electrical conversion in a vibration pressure transducer
- An oscillator and a resonator
 - A resolver and a torsion bar
 - A resistive bridge and an E-pickup
227. Which element performs the electrical conversion in a piezoresistive pressure transducer
- A torsion bar link
 - A resistive bridge
 - A resolver output
228. Several electronic pressure sensor principles exist. Which one is described as the most commonly used?
- The piezoresistive transducer
 - The vibrating transducer
 - The force compensated cell
229. Which three parts make up the vibrating transducer
- A diaphragm, a center body and an outer case
 - A base plate, a center body and a torsion bar
 - A base plate, a center body and an outer case
230. What is normally inside the reference pressure chamber of a vibrating transducer
- Usually pitot air
 - Usually ambient air
 - Usually a vacuum
231. From which material is the micromechanical pressure transducer chamber made
- Silicon material
 - Metal material
 - Pyrex material
232. What is the size of the micromechanical pressure transducer structure
- Only a few micrometres
 - Only a few centimetres
 - Only a few millimetres

- 233.** Which material is used together with the silicon in the micromechanical transducer
- Ceramic of the sintered type
 - Glass of the Pyrex type
 - Metal of the capsule type
- 234.** What is denoted by UIN and UOUT in the piezoresistive transducer
- The supply voltage and the measuring voltage
 - The measuring voltage and the supply voltage
 - The reference voltage and the bridge voltage
- 235.** In which circuit are the strain gauge resistances connected
- In a Wheatstone bridge
 - In a resolver winding
 - In a series chain
- 236.** How many strain gauges are normally fitted on the diaphragm of a piezoresistive transducer
- Four resistors
 - Eight resistors
 - Two resistors
- 237.** Which supply voltage is applied to the strain gauge bridge of a piezoresistive transducer
- 5 volts
 - 10 volts
 - 8 volts

Level 2 — understand and explain

- 238.** Pressure transducers have converted the measured pressures into electrical signals. What happens with these signals next?
- They are sent to the probe heater and transformed into the required current
 - They are sent to the computer and transformed into the required output signals
 - They are sent to the indicator and transformed into the required needle deflection
- 239.** How did the first pressure transducers work
- With metal pressure capsules, electromotively according to the follow-up principle
 - With semiconductive pressure chambers, electrically according to the bridge principle
 - With metal pressure capsules, mechanically according to the resonance principle
- 240.** From which material are the pressure chambers of the latest generation of transducers made
- Thin metallic material
 - Pyrex glass material
 - Semiconductive material
- 241.** What is stored in the integrated memory of a latest generation pressure transducer
- The reference voltage
 - The measured pressures
 - The error characteristics
- 242.** In the pressure cell type of transducer the pressure acts on one side of a metal diaphragm. What is on the other side?
- A voltage supply
 - A vacuum space
 - A pitot supply
- 243.** Which characteristic applies to the simple pressure cell transducer
- It is sensitive to pressure changes and mechanically complex
 - It is sensitive to pressure changes and electronically complex
 - It is insensitive to pressure changes and mechanically complex
- 244.** How are the instrument errors of a vibration pressure transducer handled
- They are compensated electrically
 - They are compensated by a torsion bar
 - They are compensated mechanically
- 245.** Which kind of complexity is stated for the vibration pressure transducer
- It is pneumatically complex
 - It is electronically complex
 - It is mechanically complex

- 246.** Two of the transducer types deliver a digital output signal. Which two are these?
- a. The vibration and the piezoresistive transducer
 - b. The pressure cell and the force compensated pressure cell
 - c. The pressure cell and the piezoresistive transducer
- 247.** Two of the transducer types deliver an analog output signal. Which two are these?
- a. The pressure cell and the vibration pressure transducer
 - b. The pressure cell and the force compensated pressure cell
 - c. The force compensated cell and the piezoresistive transducer
- 248.** Which output does a vibrating transducer generate
- a. An output voltage which depends on the input pressure
 - b. An output frequency which depends on the input pressure
 - c. An output frequency which depends on the reference pressure
- 249.** Which function does the outer case of a vibrating transducer have
- a. It forms the inner wall of the reference pressure chamber
 - b. It forms the outer wall of the measured pressure chamber
 - c. It forms the outer wall of the reference pressure chamber
- 250.** Which part forms the inner wall of the reference chamber
- a. The center body part
 - b. The vibrating cylinder
 - c. The base plate part
- 251.** The cylinder of a vibrating transducer starts to vibrate at its natural frequency. What has caused this?
- a. A magnetic field in the drive coil
 - b. A pressure pulse in the reference chamber
 - c. A magnetic field in the pickup coil
- 252.** On what does the natural vibrating frequency of the cylinder depend
- a. On the drive coil current and the amplifier gain
 - b. On the body itself and the environmental conditions
 - c. On the body itself and the output signal frequency
- 253.** Which task does the pickup coil have in a vibrating transducer
- a. It measures the vibrating frequency of the cylinder
 - b. It seals the reference chamber against the measured pressure
 - c. It drives the cylinder at its natural frequency
- 254.** Which physical behaviour works with the same forces as the vibrating cylinder
- a. The resonance of an aneroid capsule under constant pressure
 - b. The deflection of a Bourdon tube under internal pressure
 - c. The natural frequency of violin strings or tuning forks
- 255.** What happens when pressure acts on the silicon lid of a piezoresistive transducer
- a. The lid is dented and the capacitance in its surface changes
 - b. The lid is dented and the resistance in its surface changes
 - c. The lid vibrates and the frequency of its surface changes
- 256.** What is the output voltage of a piezoresistive transducer proportional to
- a. The supply frequency
 - b. The reference pressure
 - c. The input pressure
- 257.** A piezoresistive transducer feeds an air data computer. What is done with its output voltage before the computer uses it?
- a. It is resolved
 - b. It is rectified
 - c. It is digitized
- 258.** The micromechanical transducer has two chambers separated by the diaphragm. Which chambers are these?
- a. A chamber for input pressure and a chamber for reference pressure
 - b. A chamber for input pressure and a chamber for static pressure
 - c. A chamber for total pressure and a chamber for reference pressure
- 259.** How are the four strain gauges distributed on the base plate
- a. Three on the front side and one on the back side
 - b. All four on the front side of the plate
 - c. Two on the front side and two on the back side

260. No input pressure is applied to a piezoresistive transducer. What is the condition of the base plate and the bridge?
- The plate is straight, all resistors are equal and the bridge is balanced
 - The plate is bent, all resistors are equal and the bridge is balanced
 - The plate is straight, the front resistors are lower and the bridge is balanced
261. Which output voltage results when the bridge is balanced
- 10 volts
 - 5 volts
 - 0 volts
262. What happens to the resistances when pressure bends the base plate
- Both the front and the back resistance increase
 - The front resistance decreases and the back resistance increases
 - The front resistance increases and the back resistance decreases
263. What is the result of the resistance change when the base plate bends
- An imbalance in the bridge which causes an imbalance in the output
 - A balance in the bridge which causes a constant output
 - An imbalance in the supply which causes a drop in the input voltage
264. The resistance of a strain gauge wire follows a simple relationship. Which two factors determine it?
- The type of material and the applied supply voltage
 - The type of material and the dimension of the wire
 - The bridge balance and the dimension of the wire
265. How does the material constant of a strain gauge behave during bending
- It rises with the bending
 - It remains the same value
 - It falls with the bending
266. How does the resistance of a strain gauge change with the length of its wire
- It decreases with the length
 - It increases with the length
 - It is independent of the length
267. Which relationship gives the resistance of a strain gauge wire
- Length times cross-sectional area divided by material constant
 - Material constant times length divided by cross-sectional area
 - Material constant times cross-sectional area divided by length

Level 3 — apply and reason

268. A force compensated pressure cell is fitted instead of a plain pressure cell. Which two advantages are gained?
- It is sensitive to pressure changes and has small instrument errors
 - It is insensitive to pressure changes and needs no separate pressure cells
 - It is insensitive to pressure changes and has small instrument errors
269. One of the pressure sensing types needs no different pressure cells for static and pitot. Which type is this?
- The vibration pressure transducer
 - The plain pressure cell type
 - The force compensated pressure cell
270. A vibrating transducer is cut open. Where is the pressure that is being measured found?
- In the reference chamber outside the vibrating cylinder
 - In the pressure chamber inside the center body
 - In the pressure chamber inside the vibrating cylinder
271. What is done with the signal from the pickup coil
- It is amplified and used as output, and fed back to the reference chamber
 - It is amplified and used as output, and fed back to the drive coil
 - It is digitized and used as reference, and fed back to the drive coil
272. Why is the pickup coil signal fed back to the drive coil of a vibrating transducer?
- To keep the cylinder vibrating with its natural frequency
 - To keep the output frequency at a constant value
 - To keep the reference chamber at a constant vacuum

- 273.** Pressure is applied to a piezoresistive transducer. How do the two bridge voltages behave?
- a.** U1 increases while U2 decreases
 - b.** U1 decreases while U2 increases
 - c.** U1 and U2 both increase
- 274.** A strain gauge wire is made thicker while its length stays the same. How does its resistance change?
- a.** It is independent of the area
 - b.** It decreases with increasing area
 - c.** It increases with increasing area
- 275.** A strain gauge on the bent side of the diaphragm is stretched. Why does its resistance rise?
- a.** Because the material constant of the wire rises with the strain
 - b.** Because the wire becomes longer and its cross section smaller
 - c.** Because the wire becomes shorter and its cross section larger

Answer key

No.	Ans	Level	LTT page	Why
1	c	L1	2	The text splits classical cockpit instruments into flight monitoring and aircraft monitoring.
2	b	L1	2	The ADI is listed among the flight monitoring instruments.
3	b	L1	2	RPM indicators are engine monitoring devices, which form part of aircraft monitoring.
4	c	L1	2	The turn and slip indicator appears in the flight monitoring list; the others are aircraft monitoring devices.
5	c	L1	2	Airframe monitoring devices are given as position indicators and pressure indicators.
6	c	L1	2	The engine monitoring devices listed are RPM, exhaust gas temperature and fuel indicators.
7	c	L1	2	EIS is given as Electronic Indicating System.
8	b	L1	2	EFIS is given as Electronic Flight Instrument System.
9	c	L1	2	Flight monitoring instruments are assigned to the EFIS.
10	b	L1	2	EICAS is given as Engine Indicating and Crew Alerting System.
11	a	L1	2	ECAM is given as Electronic Centralized Aircraft Monitoring.
12	a	L1	2	CMS is given as Central Maintenance System.
13	c	L1	2	The standby devices listed are altimeter, airspeed indicator, horizon and compass.
14	c	L1	2	Earth's magnetism is one of the listed values whose change causes instruments to operate.
15	a	L1	2	The list of values names gyro stability, together with inertia and acceleration.
16	b	L2	2	Exhaust gas temperature indicators are named as engine monitoring devices under aircraft monitoring.
17	a	L2	2	Aircraft monitoring covers airframe monitoring devices and engine monitoring devices.
18	b	L2	2	The text states that aircraft monitoring instruments have different names depending on aircraft type.
19	b	L2	2	EICAS and ECAM are the names given to the aircraft monitoring instruments.
20	b	L2	2	The CMS is used for aircraft monitoring because it delivers information to maintenance personnel.
21	b	L2	2	Fluid pressure is listed separately from air pressure as a value that operates instruments.
22	a	L3	2	The standby set contains an airspeed indicator and a horizon, but no machmeter.
23	b	L1	8	The manual states the instrumentation is duplicated so that each pilot has his own set.
24	b	L1	14	Those three parameters are the ones the Basic T set is defined to present.
25	b	L1	14	The variometer is listed as the rate of climb/dive instrument of the set.
26	b	L1	14	The Basic Six list gives the airspeed indicator in knots and the altimeter in feet.
27	c	L1	14	The set lists the altimeter in feet.
28	c	L1	14	The directional gyro is named as the compass of the Basic Six.
29	a	L1	14	The list gives the artificial horizon as the aircraft attitude instrument.
30	b	L1	14	The set is defined as six essential instruments, hence the name Basic Six.
31	c	L1	16	The four instruments arranged in the T shape are the airspeed indicator, attitude indicator, altimeter and compass; turn coordination and vertical speed sit below them.
32	c	L1	16	The Basic T places the attitude indicator centrally at the top of the group.
33	b	L1	16	The altimeter occupies the position to the right of the attitude indicator.
34	a	L1	16	The airspeed indicator takes the left arm of the T.
35	b	L1	16	The compass forms the stem of the T below the attitude indicator.
36	b	L1	16	Its usual mounting position is between the windscreens, clear of the main panel.
37	a	L1	16	Those three are listed as the lower row of the Basic T.
38	b	L1	16	ADI is the Attitude Director Indicator.
39	c	L1	16	The ADI combines attitude with flight director guidance.
40	c	L1	16	HSI is the Horizontal Situation Indicator.
41	a	L1	16	PFD stands for Primary Flight Display.
42	a	L1	16	Vertical speed is the fifth Basic T parameter listed on the PFD.
43	c	L1	17	Lateral deviation is shown along the bottom, below the attitude area.
44	b	L1	17	Vertical deviation occupies the right-hand strip, opposite the airspeed scale.
45	b	L1	18	Simplification and centralisation are the two words the manual uses for the trend.
46	c	L1	19	Touch, viewing direction control and voice are the next stage shown for control input.
47	c	L1	19	Navigation is grouped onto the navigation display.
48	c	L2	14	The identical panel layout removes most of the retraining needed for blind flying on a new type.
49	c	L2	16	The HSI is the instrument that combines compass and radio navigation information.

50	c	L2	16	The PFD keeps the Basic T arrangement rather than abandoning it.
51	b	L2	17	Attitude with guidance occupies the middle of the PFD, in place of the ADI.
52	a	L2	19	The individual operating devices are drawn together into integrated central control units.
53	c	L2	19	Attitude is grouped onto the PFD and, where fitted, the HUD.
54	a	L2	19	Engine information goes to the engine/warning display and systems to the system display.
55	a	L1	20	Nitrogen makes up 78% of dry clean air at the surface.
56	b	L1	20	Oxygen makes up 21% of dry clean air at the surface.
57	c	L1	20	The remaining one per cent is the noble gas group named in the text.
58	c	L1	20	The troposphere is the layer in contact with the earth's surface.
59	a	L1	20	Weather is contained in the troposphere.
60	a	L1	20	At the poles the troposphere reaches only about 8.5 km.
61	a	L1	21	Above the troposphere and its tropopause boundary comes the stratosphere.
62	c	L1	21	The tropopause caps the troposphere.
63	b	L1	21	The stratosphere extends to about 50 km, where the stratopause lies.
64	b	L1	21	The ozone layer sits in the stratosphere, which is why that region is also called the ozonosphere.
65	c	L1	21	The exosphere is the highest layer shown.
66	c	L1	22	Flight monitoring is built on the two pressures, static and dynamic.
67	b	L1	22	Dynamic pressure is the ram pressure produced by the aircraft's motion.
68	a	L1	22	Static pressure is the atmospheric pressure surrounding the aircraft.
69	b	L1	22	Those four come from the pressures alone; TAS and SAT need temperature in addition.
70	a	L1	22	ISA sea level density is 1.225 kg per cubic metre.
71	b	L1	22	ISA mean sea level is 15 °C at 1013 mbar.
72	a	L1	22	Barometric altitude comes from static pressure only.
73	b	L1	23	Everest is still well below the tropopause, inside the troposphere.
74	b	L1	24	1013.25 hPa is the same pressure as 29.92 inches of mercury.
75	a	L1	24	15 °C corresponds to 59 °F in the ISA table.
76	a	L1	24	The ISA assumes completely dry air.
77	c	L1	24	ISA gives -56.5 °C at the 11000 m tropopause.
78	c	L1	25	760 mm Hg is the same sea level pressure as 1013.25 mb.
79	a	L1	25	ISA sea level pressure is 14.696 psi.
80	c	L1	25	The chart places the tropopause at 11 km.
81	c	L2	20	The troposphere lapse rate is 0.65 °C per 100 m of climb.
82	b	L2	21	Above the mesopause the temperature rises again through the thermosphere.
83	a	L2	22	All three listed values fall with increasing altitude.
84	b	L2	22	Temperature is the corrective value that makes TAS and SAT available.
85	b	L2	22	Calibration is referred to the ISA mean sea level values.
86	a	L2	22	The ISA is the ICAO standard atmosphere.
87	c	L2	24	The ISA exists so performance figures can be compared on a common basis.
88	b	L3	24	Identical temperatures at both levels show the lapse rate has stopped at the tropopause.
89	a	L1	28	HUD is short for Head Up Display.
90	b	L1	29	Airspeed occupies the left-hand tape of the PFD.
91	b	L1	30	The reference is entered on the altimeter's pressure scale, and the indication follows from it.
92	a	L1	30	Elevation refers to something on the ground measured from mean sea level.
93	a	L1	30	Elevation carries the setting code QFF.
94	c	L1	30	Height is measured from the ground below the aircraft.
95	b	L1	30	Height above ground is normally taken from the low range radio altimeter.
96	c	L1	30	Barometric altitude above mean sea level is flown on QNH.
97	b	L1	30	QNH is the actual pressure reduced to mean sea level.
98	c	L1	30	The standard setting is coded QNE.
99	b	L1	30	Standard setting is 1013.2 hPa.
100	b	L1	30	With 1013.2 hPa set, the indication is pressure altitude.
101	c	L1	30	Flight levels are counted from the standard 1013.25 hPa pressure surface.

102	a	L1	30	At least 1000 ft of margin is required.
103	a	L1	32	Reduced vertical separation minimum halves the vertical spacing used up high.
104	b	L1	32	Below FL 285 the separation has always been 1000 ft.
105	b	L1	32	Above FL 285 the spacing used to be doubled to 2000 ft.
106	a	L1	32	At sea level 1 hPa is worth about 30 ft.
107	c	L1	32	RVSM airspace starts above FL 290 and uses 1000 ft spacing.
108	a	L1	32	MNPS (Minimum Navigation Performance Specifications) is the specification named for RVSM approval.
109	a	L1	32	The autopilot altitude hold mode is on the list of RVSM-sensitive items.
110	b	L1	32	Two independent altimeters are required.
111	b	L1	32	Europe, America and the Pacific are the regions named.
112	a	L1	34	Training covers both flight crew and maintenance staff.
113	c	L1	34	The pitot-static sensors are replaced every 9,600 hours.
114	b	L1	34	SSEC is the correction applied for static source error.
115	c	L1	34	MNPS approval is added to the existing long range navigation approval.
116	b	L2	26	Either the instrument does the whole job or a separate air data calculator does the conversion.
117	a	L2	26	The manual attributes the differences to manufacturer and technology level.
118	b	L2	28	The HUD projects the PFD information into the pilot's forward view.
119	c	L2	30	With QFE set the reading starts from the airfield, so it is zero on the ground.
120	a	L2	30	QFE gives height above the airfield, valid while the pressure stays as set.
121	c	L2	30	QNH refers the altimeter to mean sea level, so on the ground it reads the airfield elevation.
122	a	L2	30	QNH belongs to the low level phases, below the transition altitude or level.
123	a	L2	30	The transition altitude quoted here is 5000 ft MSL.
124	c	L2	30	A common reference means all altimeters err alike, which keeps the vertical spacing valid.
125	c	L2	30	The transition altitude is the climb-out point for changing to the standard setting.
126	a	L2	30	The transition level is the descent point for changing back to QNH.
127	c	L2	32	More usable levels mean more aircraft fit into the same volume of airspace.
128	b	L2	32	Altimeters become less precise with height, so a bigger buffer was needed.
129	c	L2	32	SSE arises from where the static inlet sits in the airflow over the skin.
130	c	L2	32	The area around the sensors has to stay clean and clear so the sensed pressure is undisturbed.
131	c	L2	32	The device warns the crew that the selected altitude is being left.
132	c	L2	32	RVSM depends on the autopilot holding the selected altitude accurately.
133	a	L2	32	The transponder must report altitude taken from the altitude measurement unit.
134	c	L2	32	More available levels give better routings, saving time and fuel.
135	b	L2	33	Full implementation stretches the band up to FL 420.
136	a	L2	34	Approval is per aircraft and is applied for by the operator.
137	b	L2	34	Both the service bulletin modifications and the altitude monitoring programme are required.
138	a	L2	34	Those three are the surface waviness causes listed.
139	b	L2	34	The training list names the part numbers of the air data computer.
140	a	L2	34	Airworthiness approval comes first and opens the way to the test flight.
141	a	L2	34	Operational approval is the last step and permits use of RVSM airspace.
142	a	L2	34	Both the operation and the maintenance manuals are revised.
143	a	L3	30	Standard setting only applies above the transition altitude, so below it the altimeter stays on QNH.
144	a	L3	32	Thin air up high means pressure changes slowly with height, so one hPa covers far more feet.
145	c	L3	32	Skin damage near the static source disturbs the static pressure and so the altitude reading.
146	a	L3	34	Losing RVSM capability means the aircraft must leave RVSM airspace quickly.
147	c	L1	36	The text defines IAS as indicated airspeed.
148	c	L1	36	Glass cockpits with an ADC display CAS.
149	c	L1	37	Ptot and Pstat are the two inputs shown to the ADC.
150	b	L1	38	The manual states TAS is required for navigation.
151	b	L2	36	IAS follows dynamic pressure, which is total pressure minus static pressure.
152	a	L2	36	The bellows instrument is only true at mean sea level in standard conditions.
153	a	L2	36	CAS is IAS corrected for instrument error and static source error.

154	c	L2	36	A bellows instrument with no computer correction shows IAS.
155	a	L2	37	The ADC receives total and static pressure and applies the corrections that turn IAS into CAS.
156	a	L2	38	The formula assumes an incompressible medium.
157	a	L2	38	Compression of the air at high speed raises the total pressure, which the instrument reads as extra speed.
158	a	L2	38	The error grows with both airspeed and altitude.
159	a	L2	38	EAS is what is left when the compressibility error is taken out of CAS.
160	b	L2	38	EAS is a design and calculation value, not a cockpit indication.
161	c	L2	38	TAS is the aircraft speed relative to the surrounding air mass.
162	a	L2	39	The summary chain runs IAS to CAS to EAS to TAS.
163	b	L2	39	The last step of the chain is the density calculation.
164	a	L2	39	CAS becomes EAS when the compressibility error is corrected.
165	a	L2	39	The 15 knot compressibility error is taken off the calibrated airspeed of 293 knots to give the equivalent airspeed of 278 knots.
166	c	L3	38	The extra total pressure from compression makes the reading too high, so the correction reduces the speed.
167	b	L3	39	The error grows with altitude, so the higher aircraft has the larger error.
168	c	L1	40	The tube stands vertically and is closed at the top.
169	a	L1	40	The manual gives 760 mm as the normal mercury column height.
170	a	L1	40	The aneroid element is a thin metal capsule.
171	c	L1	40	The capsule keeps a residual pressure of about 5 mbar.
172	a	L1	40	The manual states up to eight capsules.
173	b	L1	42	The Bourdon tube arrangement turns pressure into a proportional electrical output.
174	b	L1	42	The tube has an oval cross section in the unpressurised state.
175	b	L1	42	The manual names the synchro transmitter as the usual position sensor.
176	c	L1	46	The required range is -70 °C to +50 °C.
177	c	L1	46	The manual gives 51 needle revolutions over that range.
178	a	L1	47	On the panel the variometer is the instantaneous vertical speed indicator.
179	b	L1	48	The supply is by piping and flexible tubes with screw or quick release connectors.
180	a	L1	48	The capsule stroke is only 4 to 5 mm.
181	b	L2	40	The weight of the fluid draws it down and leaves a negative pressure above it.
182	b	L2	40	Ambient static pressure on the container balances the column and holds it at a height.
183	a	L2	40	Both fluid and tube expand with temperature, so the reading must be corrected.
184	c	L2	40	Exact results need the raw reading reduced to standard conditions.
185	c	L2	40	The small residual pressure offsets temperature-driven changes in capsule elasticity.
186	b	L2	40	Stacking capsules is done to make the measurement more reliable.
187	c	L2	40	Rising pressure squeezes the capsule, so it contracts.
188	b	L2	40	The capsule stroke drives the needle by mechanical linkage.
189	a	L2	41	The aircraft aneroid altimeter senses static pressure acting on an evacuated capsule.
190	c	L2	41	The aneroid altimeter carries a temperature compensation in addition to the capsule and the indicating linkage.
191	b	L2	42	Only the curved oval tube and a position sensor at its tip are required.
192	a	L2	42	The sensor senses tip position, not pressure directly.
193	b	L2	44	The pressure acts equally over the whole tube.
194	c	L2	44	Pressure rounds the oval section into a circular one.
195	a	L2	44	Rounding the section straightens the curved tube and carries the tip outwards.
196	c	L2	44	Tip travel is proportional to the applied pressure.
197	a	L2	46	The requirement list includes insensitivity to acceleration.
198	a	L2	46	The requirements are laid down in law, so the design freedom is small.
199	c	L2	46	Direct pressure sensor instruments survive only as back-up instruments.
200	a	L2	46	A stack of thin metal diaphragm capsules is called a bellows.
201	a	L2	47	All four of these are pressure sensing instruments shown on the panel.
202	b	L2	48	The mechanism amplifies the tiny stroke into needle rotation.
203	c	L2	48	Friction in the mechanism corrupts the reading.
204	a	L2	48	Good bearings and clean mechanical construction minimise friction.
205	b	L2	48	Vibration keeps the altimeter mechanism from sticking, helping against friction.

206	a	L2	48	A one-sided spring tension takes up the gear backlash.
207	b	L2	48	Hysteresis is the difference between the rising and falling pressure characteristics.
208	a	L2	48	Long-held constant pressure makes the capsule yield slowly.
209	c	L2	48	The stroke lags in time, which is the elastic aftereffect.
210	b	L2	48	The manual states these errors act on all aneroid instruments.
211	c	L2	49	Temperature compensation in the mechanical altimeter is done with a bimetal element.
212	a	L2	49	Both capsule errors are shown as the diaphragm stroke plotted against the applied pressure.
213	c	L2	49	The pinion on the pointer axis turns the pointer.
214	c	L3	46	Such fine resolution needs the best material and fine mechanics.
215	c	L3	48	The three listed measures are special alloys, artificial ageing and particular waveforms.
216	c	L3	48	Compensation is hard because of the capsule mounting and the variable steering levers.
217	c	L3	48	Slow drift under a constant held pressure is the elastic aftereffect, not hysteresis.
218	a	L3	48	A difference between the rising and falling approach to the same pressure is hysteresis.
219	b	L3	48	The mechanical chain runs capsule, levers, shaft, cogwheel, needle.
220	c	L1	50	Transducers turn pressure into electrical signals.
221	c	L1	50	The manual names air data computers as an application.
222	c	L1	50	The newest transducers give a digital output.
223	b	L1	50	The reference is either a vacuum or 1013 mbar.
224	b	L1	51	The pressure cell with resolver gives an analog output.
225	c	L1	51	The force compensated cell uses a torsion bar and an E-pickup.
226	a	L1	51	The vibration type converts with an oscillator and a resonator.
227	b	L1	51	The piezoresistive type converts with a resistive bridge.
228	b	L1	52	The manual calls the vibrating transducer the most commonly used type.
229	c	L1	52	The three named parts are base plate, center body and outer case.
230	c	L1	52	The reference chamber usually holds a vacuum.
231	a	L1	54	The chamber is made of silicon.
232	c	L1	54	The structure measures only a few millimetres.
233	b	L1	54	The construction uses silicon with Pyrex glass.
234	a	L1	54	UIN is the supply and UOUT the measuring voltage.
235	a	L1	55	The strain gauges are wired as a Wheatstone bridge.
236	a	L1	56	Normally four strain gauges are used.
237	b	L1	57	The strain gauge bridge is supplied with 10 volts.
238	b	L2	50	The signals go to the computer, which makes the required outputs.
239	a	L2	50	Early transducers used metal capsules working electromotively on the follow-up principle.
240	c	L2	50	Modern transducer chambers are of semiconductive material.
241	c	L2	50	The built-in memory holds the error characteristics of that unit.
242	b	L2	50	The metal diaphragm of the pressure cell works against a vacuum.
243	a	L2	51	The plain pressure cell is sensitive to pressure changes and mechanically complex.
244	a	L2	51	The vibration type compensates its errors electrically.
245	b	L2	51	The vibration type is described as electronically complex.
246	a	L2	51	Both the vibration and the piezoresistive types give digital outputs.
247	b	L2	51	The two pressure cell types give analog outputs.
248	b	L2	52	Its output is a frequency that varies with the measured pressure.
249	c	L2	52	The outer case is the outer wall of the reference chamber.
250	b	L2	52	The vibrating cylinder separates the reference chamber from the measured pressure.
251	a	L2	52	The drive coil field sets the cylinder vibrating at its natural frequency.
252	b	L2	52	Both the body and the conditions around it set the natural frequency.
253	a	L2	52	The pickup coil senses how fast the cylinder is vibrating.
254	c	L2	53	The manual compares the cylinder to violin strings and tuning forks.
255	b	L2	54	Denting the lid deforms the surface resistors and changes their resistance.
256	c	L2	54	The output voltage rises in proportion to the applied pressure.
257	c	L2	54	The analogue bridge voltage is digitized for the computer.

258	a	L2	55	The diaphragm separates the input pressure chamber from the reference pressure chamber.
259	c	L2	56	Two gauges sit on the front and two on the back.
260	a	L2	56	With no pressure the plate stays straight, all resistances are equal and the bridge balances.
261	c	L2	56	Equal voltages on both sides give zero output.
262	b	L2	56	Bending lowers the front resistance and raises the back one.
263	a	L2	56	Unequal resistances unbalance the bridge and so the output.
264	b	L2	56	Resistance depends on the material constant and on the wire dimensions.
265	b	L2	56	Only the dimensions change with bending, not the material constant.
266	b	L2	56	Longer wire gives more resistance.
267	b	L2	57	The formula is R equals the material constant times length over area.
268	c	L3	51	The force compensated cell is insensitive to pressure changes and keeps instrument errors small.
269	a	L3	51	The table gives 'no different pressure cells for static and pitot' for the vibration type; both pressure cell types need separate cells.
270	c	L3	52	The measured pressure sits in the chamber inside the cylinder.
271	b	L3	52	The pickup signal serves both as the amplified output and as the feedback to the drive coil.
272	a	L3	52	The feedback loop sustains vibration at the natural frequency.
273	a	L3	56	The two halves of the bridge move in opposite directions when the plate bends.
274	b	L3	56	A larger cross section lowers the resistance.
275	b	L3	56	Stretching lengthens the wire and thins it, and both changes raise the resistance.